

Cassandra Papp

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SUMMARY

Searching for an entry-level position in AI Engineering or Data Analysis that will allow me to leverage my education and experience in Linguistics, AI, Data Management, and Natural Language Processing.

EDUCATION

2025 - 2026 MSc Speech & Language Processing at **University of Edinburgh** (Expected Nov 2026)
2022 - 2025 BA Linguistics at **University of Manchester** (First Class with Honours)

PROJECTS

Investigating Neural Whispered Speech — EveryVoice, Hifi-GAN, Remote Computing, TTS, Linux

- Investigated how perceived performance of whispered speech from a neural TTS model correlated to objective evaluation metrics, informed by linguistic descriptions of whispered speech.
- Trained a FastSpeech2-like TTS model on collected voice data, utilizing a remote connection to Linux GPU servers, modifying open source code to add an additional linguistically-informed loss term.

Qwen NLP — Qwen 2.5, Reinforcement Learning, Supervised Finetuning, CoMAT, GRPO, QA

- Utilised the Chain of Mathematically Annotated Thought method to investigate the reasoning abilities of the Qwen 2.5 model.
- Evaluated the performance of the Qwen model in both zero-shot and supervised finetuning scenarios.
- Performed Group Relative Policy Optimization using reward functions to improve model performance.

Limited Vocabulary Automatic Speech Recognition — Openfst, WFSTs, ASR, Viterbi

- Experimented with various methods of system tuning and pruning to improve ASR model accuracy and computational efficiency.
- Improved word error rate of the limited vocabulary Weighted Finite State Transducer model from 146% to 44%.

WORK EXPERIENCE

Data Fellowship — Ditchley Foundation Jul 2024 - Aug 2024

- Worked with the Neo4j graphing language to identify underrepresented company stakeholders.
- Leveraged attendee data to identify current and future trends in the green energy sector.

Hospitality Assistant — University of Manchester Jan 2023 - Jul 2025

- Managed the hospitality events unsupervised, ensuring customer satisfaction.
- Trained new hires on company procedures and organizational strategies.

SKILLS

Python	PyTorch, NLTK, NumPy, Sentence Transformers, Marian MT
Corpora	Regular Expressions, RStudio, Hugging Face, AntConc
Linguistics	Semantics, Syntax, Grammar, Typology
Other	SQL, Power BI, PHP, HTML, Linux